Infant Elfrick Gnanasusairaj

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Education

University of Wisconsin (4.0/4.0) Madison, WI
Bachelor of Aerospace Engineering + Computer Science *Sept. 2025 – Present*

Experience

SBEL Student Researcher Apr 2026 – Present
UW - Madison *Madison, WI*

- Collaborated with Simulation-Based Engineering Lab researchers to validate physics-based simulations through experimental testing. Using optimization tools like FEA, designed a custom inclining sandbox to compare simulated and real-world behavior.
- Designed and implemented hardware/software systems for scale models of machines, including lunar landers, to generate experimental data and improve simulation parity with real testing.

Design Innovation Lab AED Drone Dev Nov 2025 – Present
UW - Madison *Madison, WI*

- Worked in a research lab to improve emergency response during ski races, motivated by a recent cardiac fatality at the American Birkebeiner. Led the software development for a drone-based rescue system, in partnership with DJI, enabling rapid and reliable delivery of medical aid by overcoming the access and response-time limitations of snowmobile-based rescue.
- To make AED drone deployment accessible to non-expert users, designed a simple, user-friendly mobile application integrated with DJI's Mobile SDK (MSDK). Implemented a robust backend path-planning system that supports user-defined no-fly zones and computes optimal flight paths using the A* algorithm.
- Collaborated with the lab director on a technical paper describing the project's methodology, system design, and performance results. Contributed analysis, discussion of impact, and proposed directions for future work.

FTC Team Captain Sept 2018 – Sept 2025
Techno Maniacs *Acton, MA*

- Founded and grew my community's only FTC Robotics team, from 4 to 20+ members by securing funding, connecting with industry mentors, and creating a knowledge base.
- Led the architecture and implementation of the team's autonomous software, including localization, computer vision, state machines, and sensor feedback system while engineering hardware to maximize sensing accuracy, software reliability, and system integration.
- Led the team to a top-50 global ranking (7,000+ teams) and #1 in New England while organizing outreach initiatives that introduced robotics to over 5,000 students.

Projects

Open Source Intake | *Java, Android, State Machines, Onshape, 3D Printing, CAM, CNC* Dec 2024 – May 2025

- Designed and released a modular, open-source active intake system for the FTC Into the Deep game to lower the barrier to competitive robot design. Optimized roller geometry and flexible mounting enabled fast, reliable pickup without precise alignment across diverse drivetrains.
- Shared full CAD, build documentation, and usage guidelines to give back to the FTC community. Teams adapted and integrated components into their own systems, providing feedback that drove iterative updates and adoption by 100+ teams worldwide, including 3+ at the 2025 World Championship.

Skills [Strong] [Standard]

Languages: [Java, Python, HTML/CSS, JavaScript] [Kotlin, C++, C#, TypeScript, Bash]
Concepts: [OOP, Localization, Path Planning, PID] [Sensor Fusion, State Machines, Kinematics]
Frameworks: [React, Android, Unity, Node.js, Next.js] [ROS, Linux, Material-UI, Tailwind, Arduino]
Tools: [Git, Visual Studio, JetBrains IDEs, Onshape, 3D Printing, CNC, Fusion 360] [GCP, Blender, FEA, Isaac Sim]
Libraries: [TensorFlow, OpenCV, DJI MSDK, NumPy] [pandas, PyTorch, face detection, Matplotlib, NLTK]